

$$|x_2\rangle \text{ --- } \boxed{H^{k_1}} \text{ --- } |z_2\rangle$$

$$|x_2\rangle \text{ --- } \boxed{H^{k_1}} \text{ --- } |z_2\rangle$$

$$|x_1\rangle \text{ --- } \boxed{H^{k_2}} \text{ --- } |z_1\rangle$$

$$|x_1\rangle \text{ --- } \boxed{H^{k_2}} \text{ --- } |z_1\rangle$$

$$|x_4\rangle \text{ --- } \boxed{H^{k_3}} \text{ --- } |z_4\rangle$$

$$|x_4\rangle \text{ --- } \boxed{H^{k_3}} \text{ --- } |z_4\rangle$$

$$|x_3\rangle \text{ --- } \boxed{H^{k_4}} \text{ --- } |z_3\rangle$$

$$|x_3\rangle \text{ --- } \boxed{H^{k_4}} \text{ --- } |z_3\rangle$$